RouteVolt

Trip planning that keeps electric vehicles from running out of charge on Indian roads, starting with Tamil Nadu

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1. Problem Description

Electric vehicle (EV) adoption in India is constrained by uncertainty about charging during travel. Public charging stations are limited in number and unevenly distributed, and the energy consumed on a trip varies with distance, speed, terrain, vehicle load and weather. Drivers therefore cannot reliably judge whether the vehicle will reach its destination or the next charger. This uncertainty, known as range anxiety, discourages EV purchase and confines many owners to short urban trips.

The problem affects private EV owners making intercity journeys, commercial drivers such as taxi and delivery operators whose income depends on vehicle availability, and fleet operators who require predictable schedules. It is most pronounced on long highway corridors in Tamil Nadu, for example Chennai to Madurai and Chennai to Coimbatore, where stations are sparsely located and their working status and availability cannot be assumed.

Current navigation applications optimise for the shortest or fastest route and do not account for battery state. EV-specific tools generally display nearby charging stations but do not produce a trip plan. They do not specify at which stations to stop, how much to charge at each, or which alternative to use if a selected station is unavailable. An incorrect decision can leave a vehicle stranded on a highway, creating a safety risk and recovery costs for the driver.

The problem needs attention because it directly limits the transition to electric mobility, which supports national objectives on reducing fossil fuel dependence and transport emissions. A planning system that gives drivers a complete and reliable plan, including contingencies, addresses a barrier that expansion of charging infrastructure alone does not remove.

2. Proposed Solution

RouteVolt is a journey planner for EVs. The driver enters a start point, a destination and the current battery level for their vehicle. RouteVolt returns a complete plan: the route, the charging stops, how much to charge at each, the expected arrival battery, and backup stations in case a primary stop fails.

How it works. The system has five parts that run one after another.

- **Road network and routing.** We build a graph of the road network between the origin and destination. Ola Maps supplies routing that reflects Indian road conditions, OSRM provides the distance matrix between points, and Google Maps is the backup if either source fails. Dijkstra's

algorithm then finds the best path over the graph. Because the cost of an edge is energy and time rather than distance alone, the best route for an EV may differ from the shortest one.

- **Energy prediction.** A Gradient Boosting model estimates the energy each road segment will consume, using inputs such as segment length, speed profile and the vehicle's battery size. This replaces a flat 'x km per percent' rule of thumb.
- **Battery simulation.** RouteVolt walks through the trip stop by stop and tracks the battery level, so it can tell exactly where charge would drop below a safe floor. We keep a safety margin so that the plan does not assume a perfect drive.
- **Charging station selection.** At each point where a charge is needed, the planner picks stations that fit the battery state and route, and caps charging at about 80%, since going beyond that is slow and wastes the driver's time. For every primary stop it also keeps ranked backups, so a closed or crowded station does not break the trip.
- **Capacity-aware allocation.** We do not yet have historical occupancy data from real stations, so we do not claim to predict queues. Instead the planner accounts for each station's capacity and spreads suggestions so many drivers are not pointed to the same charger.

Difference from existing tools. ABRP is a general-purpose global planner that is not built around Indian roads and charger networks, and live charger availability is a premium feature. Tata Power EZ Charge is a charging network app whose Trip Planner offers a fastest route or a route with the most stations within a detour limit set by the user; the choice of stops is left to the user, and the app does not describe how battery use along the route is estimated. RouteVolt differs in three ways. It predicts energy use for each road segment and simulates the battery stop by stop, so the plan states the charge needed at every stop. It attaches ranked backup stations to every stop in advance, instead of rerouting after a failure. It allocates stops with station capacity in mind, so that drivers are not all directed to the same charger. RouteVolt does not yet use live charger data; integrating it is part of the implementation plan below. This comparison is based on the products' public descriptions as of October 2026.

Current status and honest limits. RouteVolt is a working prototype. Its first version uses a Tamil Nadu charger dataset, and the planner itself is designed to extend across India. The energy model is trained on synthetic data, because a real labelled dataset of EV trips was not available to us. That makes its very high accuracy score less meaningful than it looks, since synthetic energy use is mostly driven by distance. The next step is to validate it against real vehicle logs. Live station availability is also not connected yet. We would rather state these limits plainly than overstate what the prototype can do.

Implementation plan.

- **Data and validation.** Validate the energy model on real trip data from partner drivers or fleet operators, and integrate live charger status from station operators or open data.
- **Pilot.** Run a pilot with a small group of EV owners on two or three Tamil Nadu corridors and compare predicted against actual arrival battery.
- **In-car delivery.** Deliver RouteVolt through Apple CarPlay and Android Auto, so the plan, the next charging stop and the backups appear on the car's display while driving, without the driver handling a phone. A mobile app will serve for trip planning before the journey.
- **Expansion.** Add road and station data for further states, starting with neighbouring southern states.

3. Expected Impact

Who benefits. Individual EV owners get confidence on longer trips. Commercial drivers and small fleets get fewer breakdowns and more predictable schedules. Charging station operators benefit when

demand is spread more evenly instead of piling up at a few stations. Society benefits if easier EV use encourages more people to switch away from petrol and diesel.

What improves. The main improvement is trust: a driver who can see the full plan, with backups, is less likely to avoid long trips or avoid EVs altogether. Fewer stranded vehicles means fewer roadside emergencies. Better use of existing chargers means the same infrastructure serves more people before new stations are built. We will measure success through the gap between predicted and actual arrival battery, the share of trips completed without an unplanned stop, and driver feedback in pilots. We are not claiming specific figures before we have tested them.

How it can be implemented. The software runs on open road data and standard cloud hosting, so the cost to start is low. It does not need new hardware. Pilots can begin with fleet operators, who have clear routes and a direct reason to adopt it, and then move to individual users.

Scaling. We plan to take RouteVolt across India, not only Tamil Nadu. The Tamil Nadu charger dataset is used for the first prototype only, and nothing in the planner depends on it. Extending to another state needs that state's road data and station list, and vehicle profiles can be added for more EV models. We would expand corridor by corridor, starting with neighbouring southern states. Over time the same planner could support state transport agencies, charging networks and EV rental services, and its trip data could help planners decide where new chargers are most needed.